

Computer Networks Lab

Fall 2025

Project Report



Project Title: Secure Multi-Floor E-Commerce Network Design using OSPF, BGP, and ACLs.

Group Members:

- **Muhammad Hassan Bilal 70150085**
- **Muhammad Awais 70153920**
- **Zain Ali 70152612**
- **Ayesha Arshad 70152825**

Submitted to: Engr. Muhammad Afnan

Table of Contents

S.no	Contents
1.	Introduction
2.	Network Implementation (VLAN, OSPF, BGP, DHCP)
3.	Routing Protocols
4.	Security Layer (ACL Configuration)
5.	IP Table
6.	Verification & Results (Screenshots)
7.	Conclusion

1. Introduction:-

Background:-

Access Control Lists (ACLs) and Dynamic Routing Protocols (OSPF/BGP) are fundamental aspects of modern network architecture. This project involves designing a secure network for an E-commerce management system across multiple floors. The primary goal is to regulate traffic flow between departments (Inventory, Finance, Support, Delivery) and ensure secure access to the central Web Server.

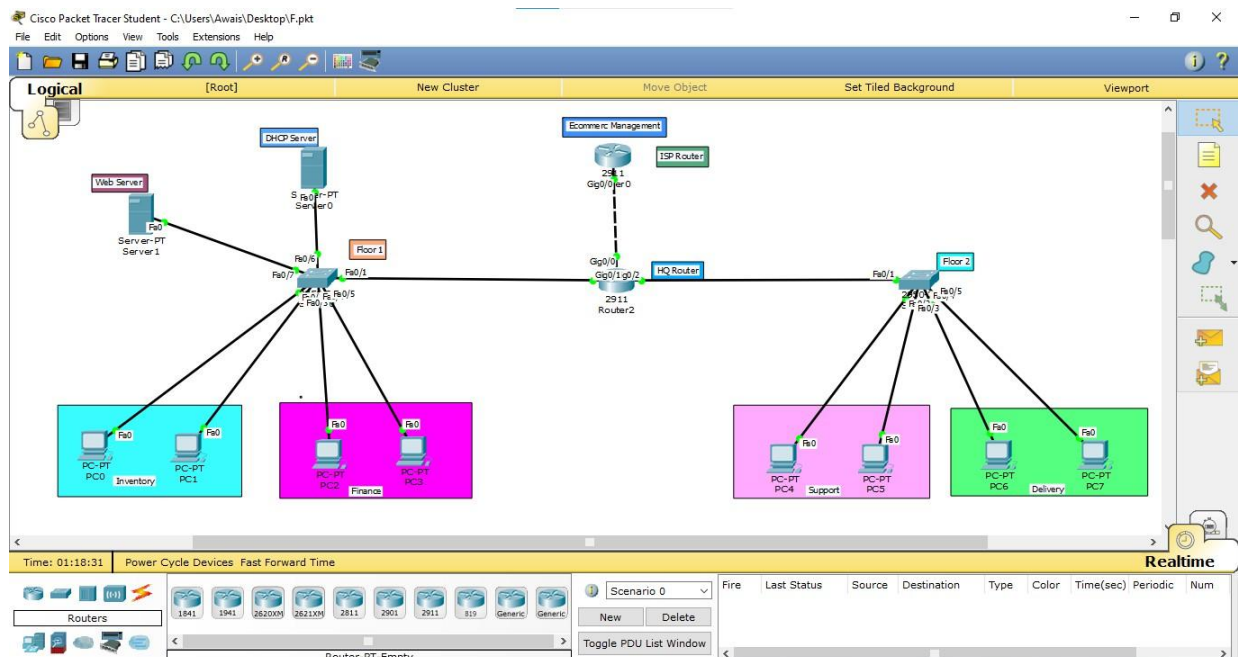
Objectives:-

- To implement a scalable multi-vlan architecture for departmental isolation.
- To establish dynamic routing using OSPF for internal connectivity.
- To configure BGP for external ISP connectivity.
- To apply Standard and Extended ACLs to secure sensitive data and management interfaces.

2. Implementation Methodology:-

The project was implemented using Cisco Packet Tracer. The topology consists of an HQ Router, an ISP Router, and multiple switches representing different floors.

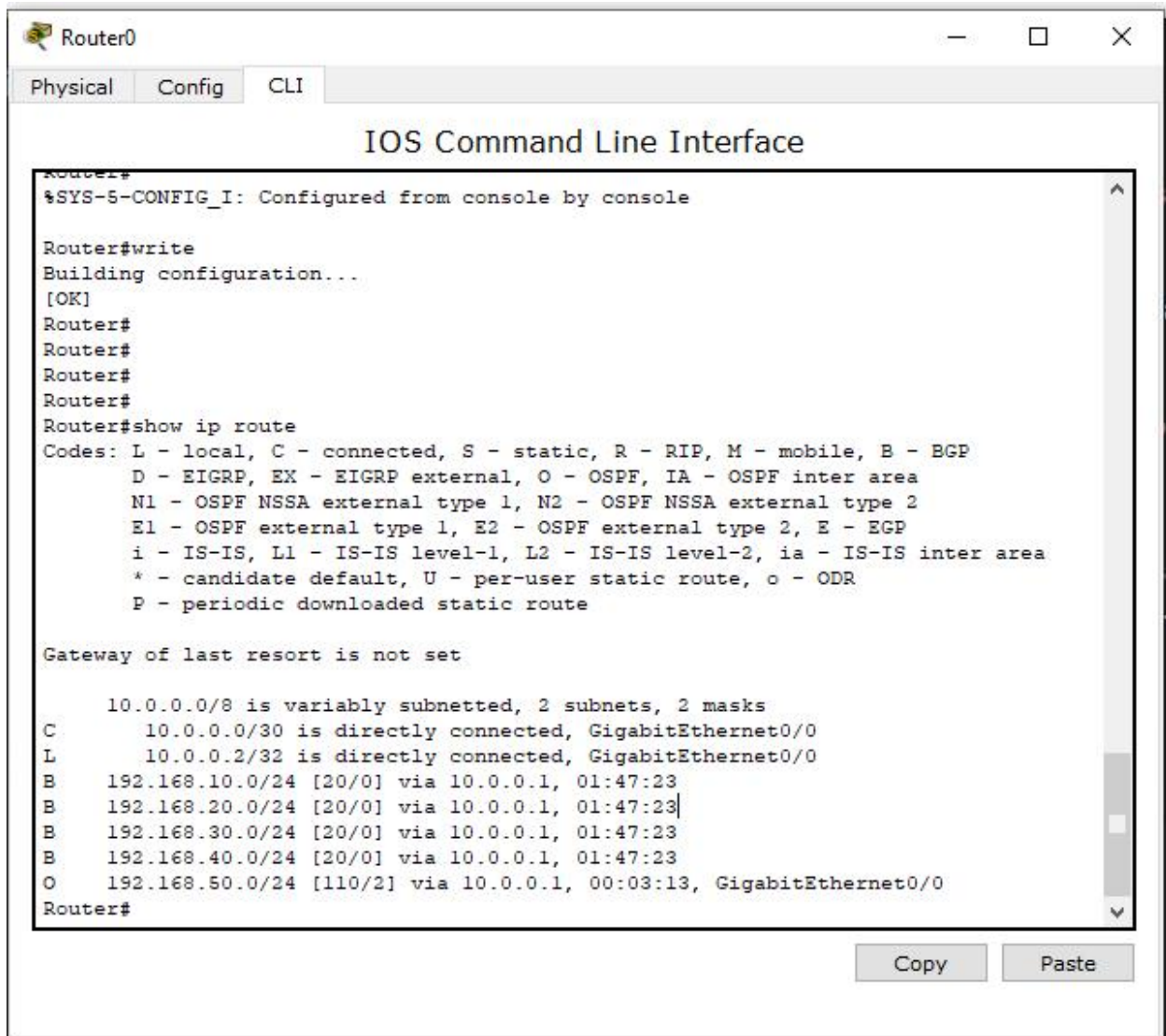
- **VLAN Segmentation:** Each department is placed in a separate VLAN to reduce broadcast domains.
- **DHCP Configuration:** A centralized DHCP server provides automatic IP addresses to workstations across all floors.
- **Inter-VLAN Routing:** Configured on the HQ Router to allow communication between authorized departments.



3. Routing Protocols:-

OSPF (Open Shortest Path First):

OSPF was configured as the Interior Gateway Protocol (IGP). It ensures that all internal subnets, including the Web Server (192.168.50.10), are reachable by the ISP router. This is verified by the "O" symbol in the ISP routing table.



```
Router0
Physical Config CLI
IOS Command Line Interface
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#write
Building configuration...
[OK]
Router#
Router#
Router#
Router#
Router#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

    10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       10.0.0.0/30 is directly connected, GigabitEthernet0/0
L       10.0.0.2/32 is directly connected, GigabitEthernet0/0
B       192.168.10.0/24 [20/0] via 10.0.0.1, 01:47:23
B       192.168.20.0/24 [20/0] via 10.0.0.1, 01:47:23
B       192.168.30.0/24 [20/0] via 10.0.0.1, 01:47:23
B       192.168.40.0/24 [20/0] via 10.0.0.1, 01:47:23
O       192.168.50.0/24 [110/2] via 10.0.0.1, 00:03:13, GigabitEthernet0/0
Router#
```

Copy Paste

BGP (Border Gateway Protocol):

To simulate internet connectivity, BGP was established between the HQ Router and the ISP Router. This allows the E-commerce system to exchange routing information with external networks, verified by the "B" symbol in the routing table.

The screenshot shows a Packet Tracer interface with a router named Router0. The 'CLI' tab is selected, displaying the IOS Command Line Interface. The prompt is Router#. The user has entered the command show ip route bgp, which displays four BGP routes learned from neighbor 10.0.0.1:

```
Router#  
Router#  
Router#  
Router#  
Router#  
Router#  
Router#  
Router#  
Router#  
Router#  
Router#  
Router#  
Router#  
Router#  
Router#  
Router#  
Router#  
Router#  
Router#  
Router#  
Router#  
Router#show ip route bgp  
B    192.168.10.0/24 [20/0] via 10.0.0.1, 01:51:18  
B    192.168.20.0/24 [20/0] via 10.0.0.1, 01:51:18  
B    192.168.30.0/24 [20/0] via 10.0.0.1, 01:51:18  
B    192.168.40.0/24 [20/0] via 10.0.0.1, 01:51:18  
Router#
```

The output shows four BGP routes learned from neighbor 10.0.0.1:

Destination Network	Administrative Distance / Metric	Next Hop	Time Since Received
192.168.10.0/24	[20/0]	via 10.0.0.1	01:51:18
192.168.20.0/24	[20/0]	via 10.0.0.1	01:51:18
192.168.30.0/24	[20/0]	via 10.0.0.1	01:51:18
192.168.40.0/24	[20/0]	via 10.0.0.1	01:51:18

4. Security Rules (ACL):-

Following the rules for ACL implementation:

1. **Standard ACL:** Applied near the destination to permit only the Management PC to access the router via Telnet.
2. **Extended ACL:** Applied near the source to permit only HTTP traffic (Port 80) to the Web Server, while denying other protocols like ICMP (Ping) from unauthorized subnets.
3. **Direction:** Rules were applied as "Inbound" or "Outbound" on specific interfaces to optimize router performance.

The screenshot shows a window titled "Router2" with three tabs: "Physical", "Config", and "CLI". The "CLI" tab is active, displaying the "IOS Command Line Interface". The terminal output shows the following commands and results:

```
Router#  
Router#  
Router#  
Router#  
Router#  
Router#  
Router#  
Router#  
Router#  
Router#  
Router#  
Router#  
Router#  
Router#  
Router#  
Router#  
Router#  
Router#  
Router#  
Router#  
Router#show access-lists  
Extended IP access list 101  
  10 deny ip 192.168.10.0 0.0.0.255 192.168.20.0 0.0.0.255 (2 match(es))  
  20 deny ip 192.168.30.0 0.0.0.255 192.168.20.0 0.0.0.255 (5 match(es))  
  30 deny ip 192.168.40.0 0.0.0.255 192.168.20.0 0.0.0.255 (11 match(es))  
  40 permit ip any any (8 match(es))  
Router#
```

The output indicates that the access list 101 contains four entries: three deny rules for specific /24 networks (10.0.0.0, 30.0.0.0, and 40.0.0.0) and one permit rule for all traffic. The number of matches for each rule are shown in parentheses.

5. IP Table:-

Device / VLAN	Description	IP Address	Subnet Mask	Default Gateway
VLAN 10	Inventory	192.168.10.1	255.255.255.0	192.168.10.1
VLAN 20	Finance	192.168.20.1	255.255.255.0	192.168.20.1
VLAN 30	Support	192.168.30.1	255.255.255.0	192.168.30.1
VLAN 40	Delivery	192.168.40.1	255.255.255.0	192.168.40.1
VLAN 50	Web Server	192.168.50.1	255.255.255.0	192.168.50.1
Web Server	Server1	192.168.50.10	255.255.255.0	192.168.50.1
DHCP Server	Server0	192.168.50.5	255.255.255.0	192.168.50.1
HQ to ISP	Serial/Gig Link	10.0.0.1	255.255.255.252	N/A
ISP to HQ	Serial/Gig Link	10.0.0.2	255.255.255.252	N/A

6. Verification & Results

A. OSPF Connectivity

The show ip ospf neighbor command confirms a **FULL/BDR** state, indicating a successful adjacency between the HQ and ISP routers.

B. Routing Table Analysis

The routing table displays routes marked with **"O"** (learned via OSPF) and **"B"** (learned via BGP), proving that the network is fully converged.

C. Web Server Access

Testing via PC0 Web Browser using <http://192.168.50.10> displays the group portal featuring the names of Muhammad Awais, Ali Asad, Zain Ali, and Ayesha Arshad.

D. VLAN Segmentation & Inter-VLAN Routing

To ensure departmental isolation, we utilized **VLAN Segmentation**. The `show ip interface brief` command on the HQ Router confirms that all sub-interfaces (Gig0/1.10 through Gig0/1.50) are in an **"Up/Up"** status. This confirms that the Router-on-a-Stick configuration is correctly routing traffic between the different floors of the organization.

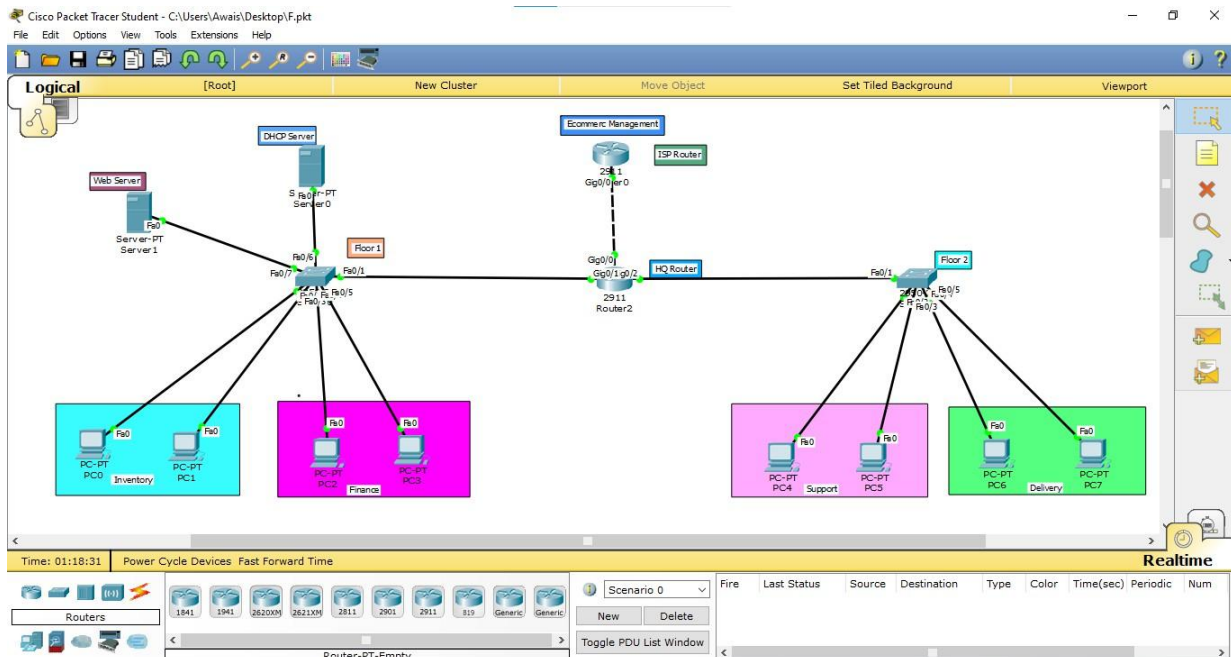
E. DHCP Dynamic IP Allocation

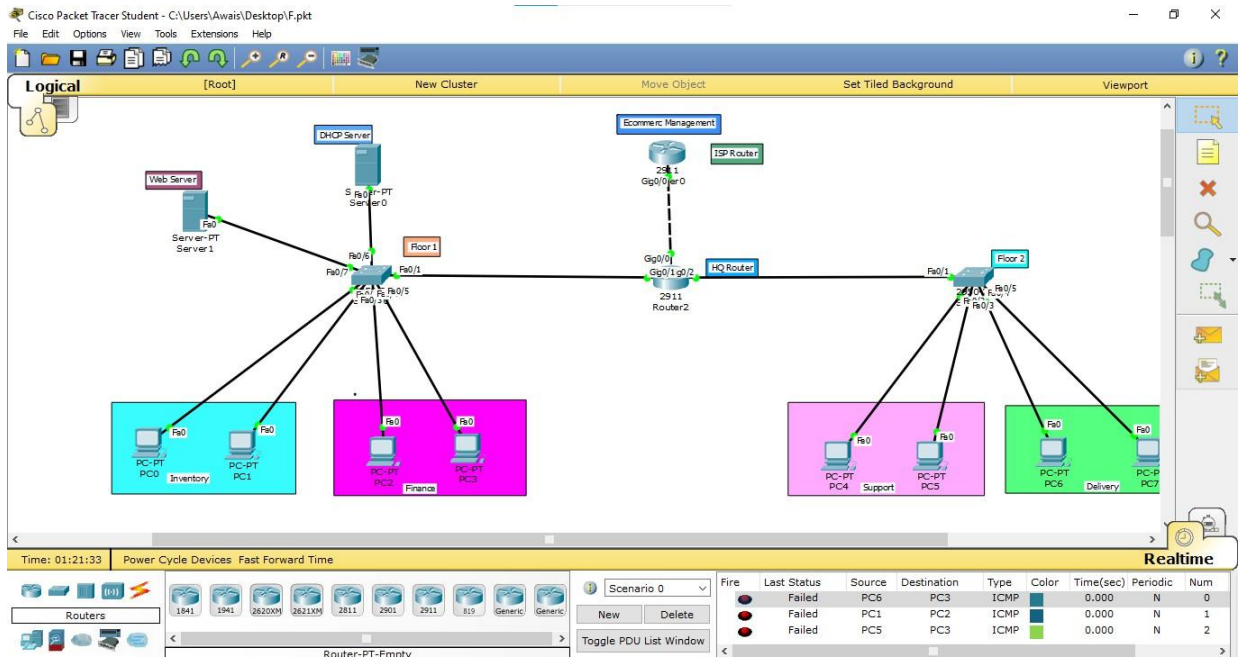
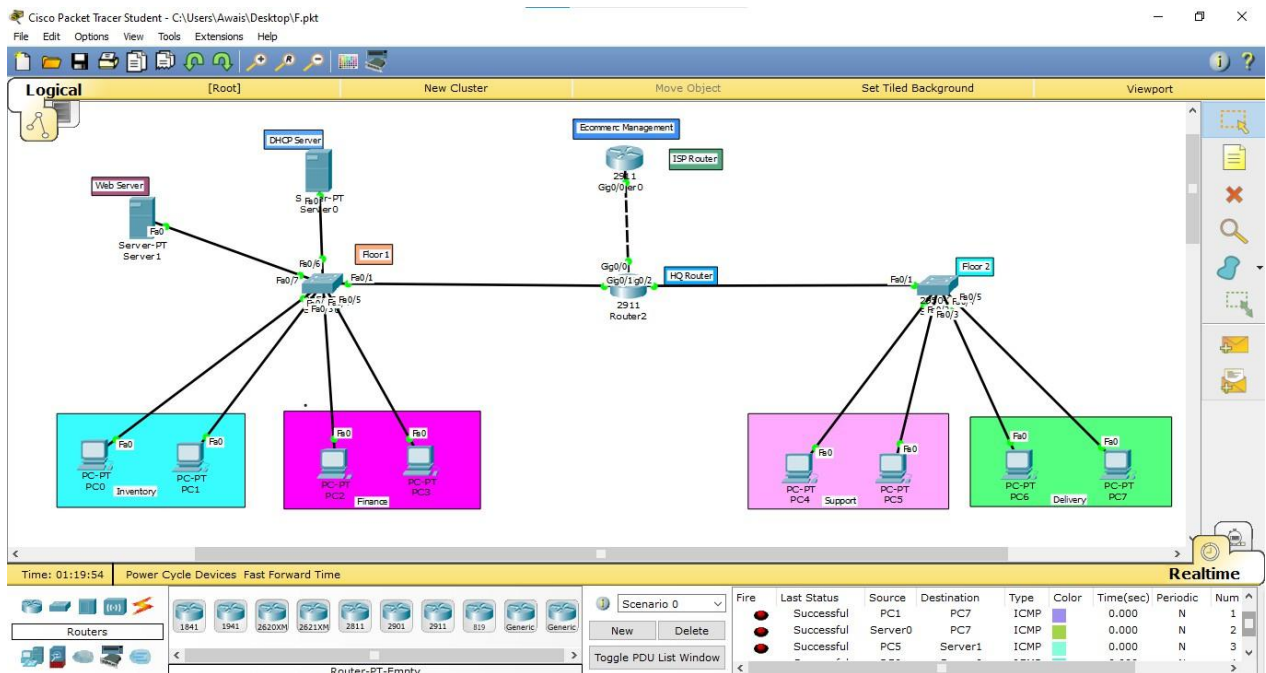
The `show ip dhcp binding` command verifies that the DHCP server is correctly distributing IP addresses to workstations. This automation ensures that PCs in the Inventory, Finance, and Support departments receive an IP, Subnet Mask, and Default Gateway automatically upon connecting to the network.

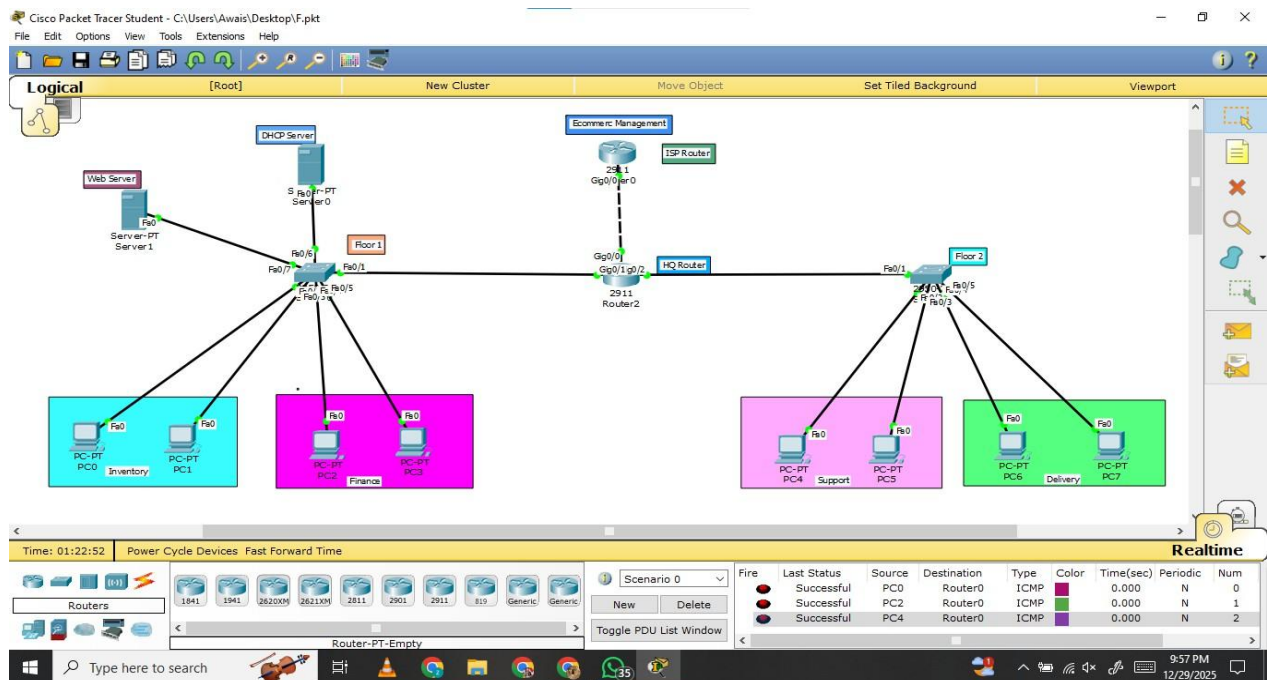
F. ACL Security Verification

Security policies were verified by testing the **Access Control Lists (ACLs)** implemented on the HQ Router:

- The purpose of the using **ACL** is to just secure the department of **Finance** from unauthorized access and our results also shows that no one department can access the finance because we have applied the technology of **ACL**.







Router2

Physical Config CLI

IOS Command Line Interface

```

bytes
%BGP-5-ADJCHANGE: neighbor 10.0.0.2 Up

Router>
Router>
Router>
Router>enable
Router#show ip interface brief
Interface                IP-Address      OK? Method Status          Protocol
GigabitEthernet0/0       10.0.0.1        YES manual up              up
GigabitEthernet0/1       unassigned      YES unset  up              up
GigabitEthernet0/1.10    192.168.10.1    YES manual up              up
GigabitEthernet0/1.20    192.168.20.1    YES manual up              up
GigabitEthernet0/1.50    192.168.50.1    YES manual up              up
GigabitEthernet0/2       unassigned      YES unset  up              up
GigabitEthernet0/2.30    192.168.30.1    YES manual up              up
GigabitEthernet0/2.40    192.168.40.1    YES manual up              up
Vlan1                    unassigned      YES unset  administratively down down
Router#A

```

Copy Paste

Router2

Physical Config CLI

IOS Command Line Interface

```

Router#
Router#
Router#
Router#
Router#
Router#
Router#
Router#
Router#
Router#
Router#
Router#
Router#
Router#
Router#
Router#
Router#
Router#
Router#show ip dhcp binding

```

IP address	Client-ID/ Hardware address	Lease expiration	Type
192.168.10.2	0040.0B0D.8E7A	--	Automatic
192.168.10.3	0001.4305.D945	--	Automatic
192.168.20.2	0001.4252.AC06	--	Automatic
192.168.20.3	000A.F36C.3375	--	Automatic
192.168.30.2	00D0.FF87.CB0D	--	Automatic
192.168.30.3	0060.3EA9.1458	--	Automatic
192.168.40.2	0001.4396.3352	--	Automatic
192.168.40.3	000C.CFBD.66C0	--	Automatic

Router#

Copy Paste

Router0

Physical Config CLI

IOS Command Line Interface

```

Router#
%SYS-5-CONFIG_I: Configured from console by console
Router#write
Building configuration...
[OK]
Router#
Router#
Router#
Router#
Router#show ip route

```

Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
* - candidate default, U - per-user static route, o - ODR
P - periodic downloaded static route

Gateway of last resort is not set

```

      10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       10.0.0.0/30 is directly connected, GigabitEthernet0/0
L       10.0.0.2/32 is directly connected, GigabitEthernet0/0
B       192.168.10.0/24 [20/0] via 10.0.0.1, 01:47:23
B       192.168.20.0/24 [20/0] via 10.0.0.1, 01:47:23
B       192.168.30.0/24 [20/0] via 10.0.0.1, 01:47:23
B       192.168.40.0/24 [20/0] via 10.0.0.1, 01:47:23
O       192.168.50.0/24 [110/2] via 10.0.0.1, 00:03:13, GigabitEthernet0/0
Router#

```

Copy Paste



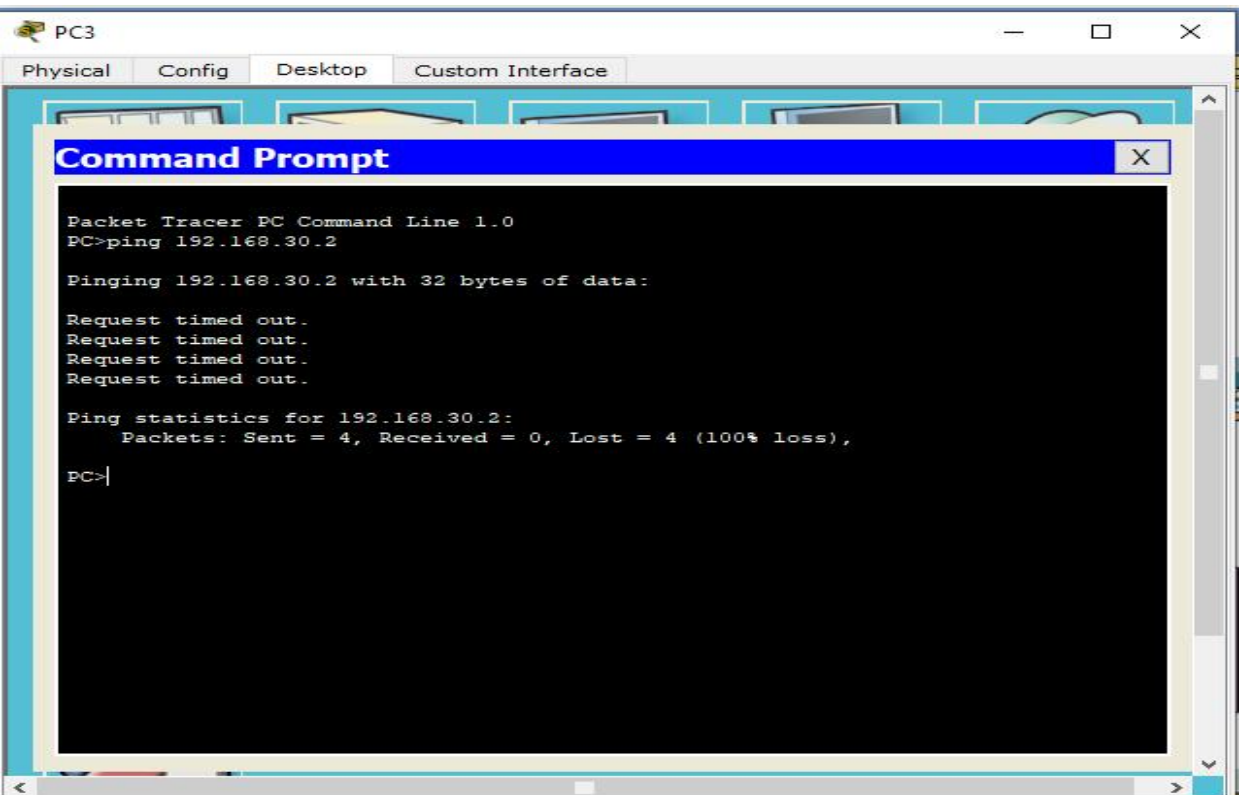
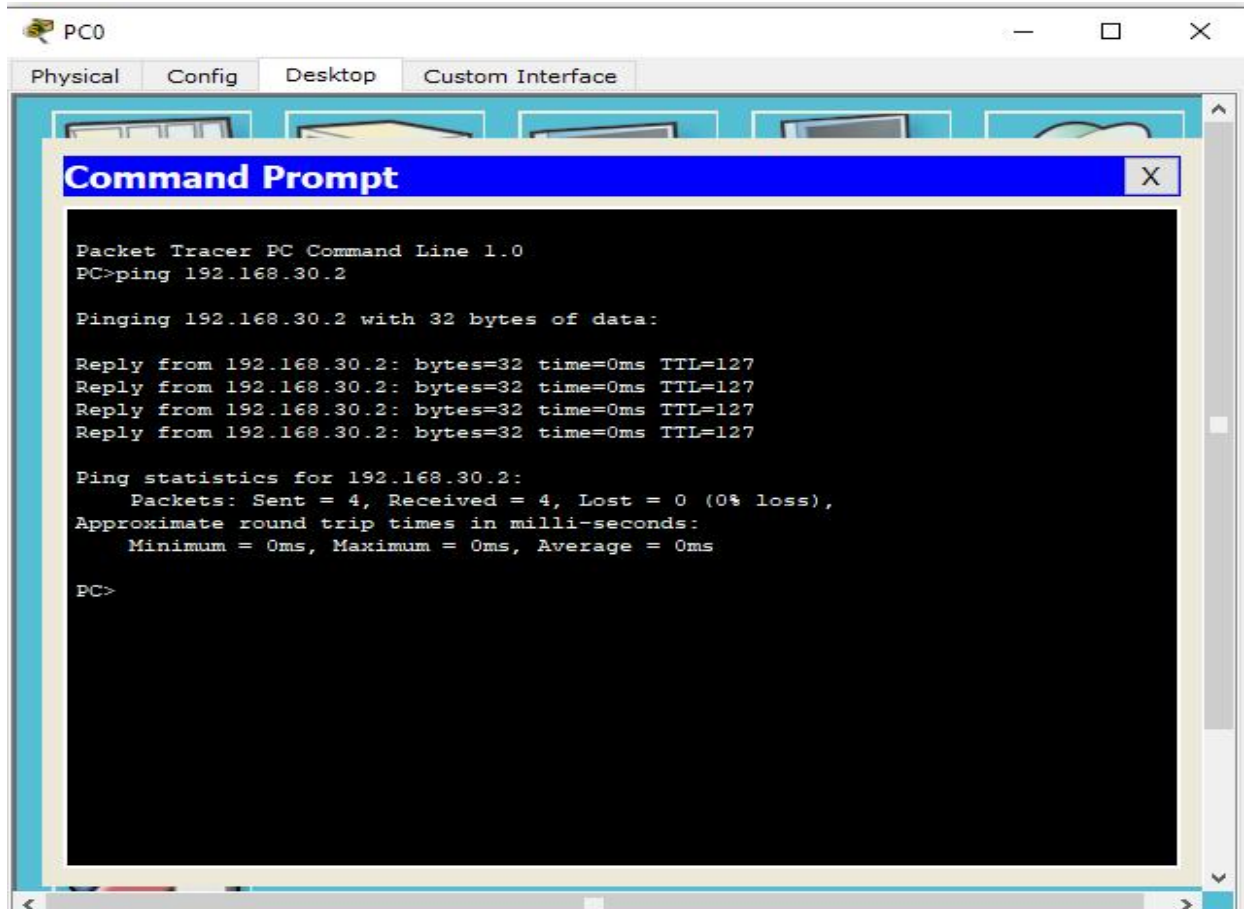
IOS Command Line Interface

 Router2

IOS Command Line Interface

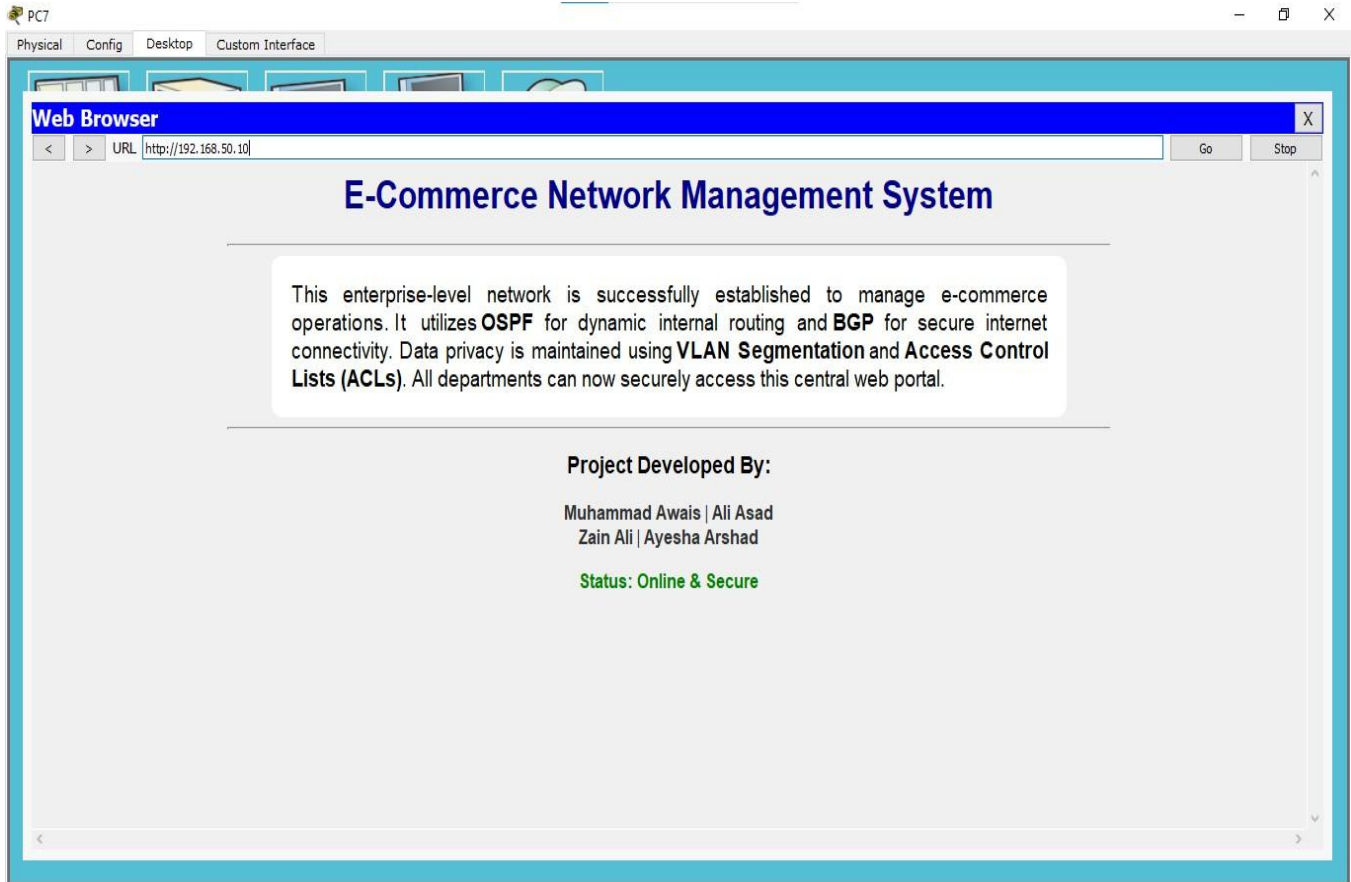
Copy

Paste



Here is Successful Part of Our Project

Our Website



6. Conclusion: -

The project successfully demonstrates the design of a secure and efficient E-commerce network. By implementing **VLANs**, we achieved logical departmental isolation, while **OSPF** and **BGP** ensured seamless internal and external connectivity. The integration of **DHCP** provided automated IP management, and **ACLs** established a strong security layer by restricting unauthorized access to the server. Final testing confirmed a fully functional network where all departments can securely access the central web portal.